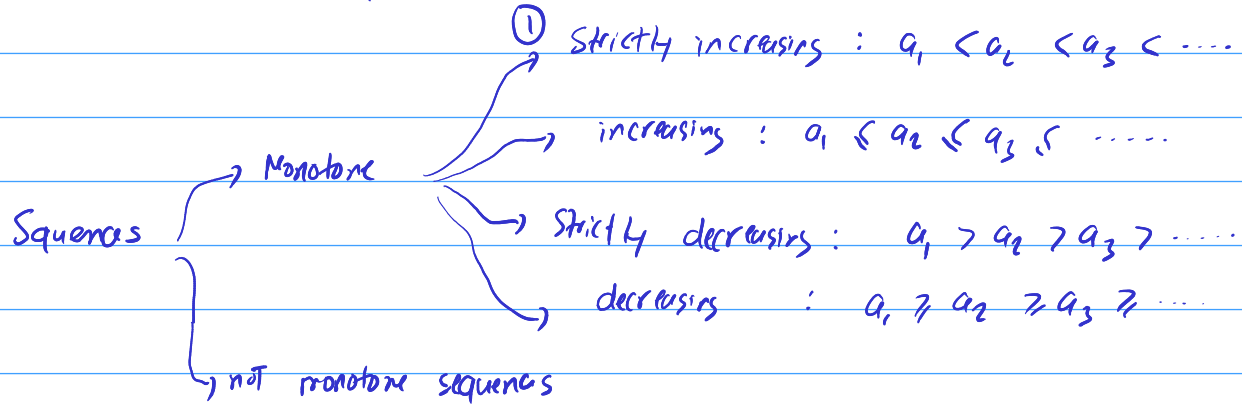
9.2 Monotone Sequences



① Strictly increasing : $a_n = n$

$$\underline{\text{OR}} \qquad \{n\}_{n=1}^{\infty}$$

$$= 1, 2, 3, 4, 5 \ldots$$

② Strictly decreasing : $a_n = -n$

$$a_n = \frac{1}{n}$$

$$= 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \ldots$$

③ not monotone sequence :

$$a_n = \sin(n)$$

Theorem :

If a_n is (eventually) increasing, then

① a_n is convergent if $a_n \leq M$ for some constant M and $\lim_{n \to \infty} a_n = L \leq M$

② $\lim a_n = \infty$ otherwise

M is an upper bound of a_n for case ①

Similarly :

If a_n is decreasing and has a lower bound N then a_n is convergent and $\lim a_n \geq N$.

If a_n is decreasing and has no lower bound then $\lim a_n = -\infty$

DIFFERENCE BETWEEN SUCCESSIVE TERMS	RATIO OF SUCCESSIVE TERMS	CONCLUSION
$a_{n+1} - a_n > 0$	$a_{n+1}/a_n > 1$	Strictly increasing
$a_{n+1} - a_n < 0$	$a_{n+1}/a_n < 1$	Strictly decreasing
$a_{n+1} - a_n \geq 0$	$a_{n+1}/a_n \geq 1$	Increasing
$a_{n+1} - a_n \leq 0$	$a_{n+1}/a_n \leq 1$	Decreasing

$a_n > 0$ is required

Example :

Show that $a_n = \frac{1}{n}$ is strictly decreasing

① use difference test:

$$a_{n+1} - a_n < 0$$

$$\Leftrightarrow \frac{1}{n+1} - \frac{1}{n} < 0$$

$$\Leftrightarrow \frac{n - (n+1)}{n(n+1)} < 0$$

$$\Leftrightarrow \frac{-1}{n(n+1)} < 0$$

This is true b/c the top is (−) and bottom (+

② Ratio test (notice that $a_n > 0$)

$$\frac{a_{n+1}}{a_n} < 1$$

$$\Leftrightarrow \frac{\frac{1}{n+1}}{\frac{1}{n}} < 1$$

$$\Leftrightarrow \frac{n}{n+1} < 1$$

$$\Leftrightarrow \frac{n}{n+1} - 1 < 0$$

$$\Leftrightarrow \frac{n}{n+1} - \frac{n+1}{n+1} < 0$$

$$\Leftrightarrow \frac{n-(n+1)}{n+1} < 0$$

$$\Leftrightarrow \frac{-1}{n+1} < 0$$

True. This means a_n is decreasing strictly

$a_n = n - 3^n$

$$a_{n+1} = (n+1) - 3^{n+1}$$

$$\Rightarrow a_{n+1} - a_n = (n+1) - 3^{n+1} - (n - 3^n)$$

$$= n + 1 - 3^{n+1} - n + 3^n$$

$$= -3^{n+1} + 3^n + 1$$

$$= -3^n \cdot 3^1 + 3^n + 1$$

$$= 3^n(-3 + 1) + 1$$

$$= 3^n \cdot (-2) + 1 < 0$$

$a_n = \dfrac{3^n}{2 + 3^n}$

$$a_{n+1} = \frac{3^{n+1}}{2 + 3^{n+1}}$$

$$\Rightarrow \frac{a_{n+1}}{a_n} = \frac{3^{n+1}}{2 + 3^{n+1}} : \frac{3^n}{2 + 3^n}$$

$$= \frac{3^n \cdot 3}{2+3^{n+1}} \cdot \frac{2+3^n}{3^n}$$

$$= \frac{3 \cdot (2+3^n)}{2+3^{n+1}}$$

$$= \frac{6+3\cdot 3^n}{2+3^{n+1}}$$

$$= \frac{6+3^{n+1}}{2+3^{n+1}} > 1$$

b/c $6 + 3^{n+1} > 2 + 3^{n+1}$

$\Rightarrow$ G_n is increasing.